ADAGIO GROUP

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Reply to responses on paper 1

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Sat, Sep 26, 2026 at 10:24 AM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>

Ben,

My assessment is favourable: both material findings are resolved, and resolved in the direction both reports asked for. Amendments remain, one of them material, and they are set out below.

Before any of that I owe you corrections to my own report and the material I failed to send.

1. The report's defects, and what I owed you.

You were right on all three counts, and I have found two more.

The broken cross-reference in Question 5 points at material that did not survive into the report I sent you. What it was meant to carry were two strengthenings of the Section 6 prediction (the one-sided form of the test, and a criterion separating classical memory from quantum indivisibility) together with the sampling floor.
You implemented the first two independently, by way of other findings, so the question turned out not to be load-bearing. The third was genuinely missing and is below. The "verification section" referred to twice does not exist in the delivered report; its counterexample is also below. And the header date is wrong: the evaluation was completed after the 7 September deposit, and the 4 September date is a carry-over from the other reports in that batch. I am issuing a corrected report.

Two further defects are mine and they are of substance, not form,
because the deposit adopted formulations my report prescribed and the prescriptions were not right.

Finding 5 recommended the reading on which "the sharp surface itself carries no algebra." Section 4 now concludes that the slab carries the determination "and the surface does not." That does not follow. The Haag–Kastler net is indexed by open regions and a sharp Cauchy surface has empty interior, so the absence of an algebra is a feature of the indexing, not a theorem that the surface fails to determine (and the axiom itself says that the algebra of *every* neighbourhood, however thin, is already the algebra of the whole spacetime, which is an initial-value structure). My formulation invited the overreach; the printed sentence is still yours to fix, and the fix is to drop the final clause.

Question 1 asked for "well-posed evolution from admissible initial data."
Definition 1(a) says "well posed" and then states existence, uniqueness and determination by $x(t_0)$, without continuous dependence on the data, which is part of the standard sense.
Either add it or say "uniquely solvable."

The counterexample. Take the generator

Q = [[−1.1, 0.7, 0.4], [0.5, −1.2, 0.7], [0.3, 0.6, −0.9]], π = (0.264, 0.348, 0.388),

and let λ merge the second and third states. The partition is not strongly lumpable in the Kemeny–Snell sense: the aggregated rates into the first group from the two states of the second are 0.5 and 0.3. The two-time family of the lumped process, started in π, has at the triple (0, 0.9, 1.8) a Chapman–Kolmogorov defect of spectral norm 0.0024979, independent of δ because the limit object does not involve the window. So it is a Markov process observed through a non-lumpable function of its state whose defect does not vanish under refinement. One convention note, since both appear at once: Q is written row-wise, with rows summing to zero, while the defect matrix is naturally displayed in your column convention (row-wise it is the transpose, with the same spectral norm). The example corroborates the second half of your Lemma 1 rather than challenging it.

The sampling floor. I give this as a conditional heuristic, not as a consequence of the lemma, and the distinction matters for the reason set out in item 3 below.
The lemma bounds from above and supplies no lower bound on the signal, no nonzero linear coefficient, no estimator variance, no dependence on small conditioning probabilities, and no correction for temporal dependence or multiple comparisons; the signal may be exactly zero or o(δ). Conditionally, then: if in addition $\|\Delta_\delta\| \sim a\, q^*\delta$ with $a > 0$ and the estimator's typical error is $\sigma_{eff}/\sqrt{N}$, the signal-to-noise balance suggests $\delta_{min} \asymp z\, \sigma_{eff} /(a\, q^* \sqrt{N})$, to be calibrated per model by simulation or bootstrap.
The barrier itself is real and I met it: with about $2.4\cdot10^5$ trajectories the sampling error is around $2\cdot10^{-3}$, and at δ = 0.01 the estimated defect of the chain I examined fell below that, so no order was resolvable without widening the range of δ.
Section 6 should require a protocol to declare such a floor, calibrated rather than derived.

2. What is resolved.

Finding 6 is the change that reorganizes the paper, and Definition 1 with the hierarchy paragraph is exactly what the question asked for: the two senses defined separately,
Markov ⇒ composition with the converse failing by Feller, composition as a necessary condition so that its failure refutes completeness and its holding does not establish it, and a statement of which sense each section concerns. The one-sided test follows.
I want to single out one sentence you added that neither report asked for and that answers Sara's principal objection directly: that the presupposition concerns the completeness of the state at a point, "not its existence, which is a matter of definition and is not in dispute here."

Finding 2 is resolved by naming the operation in force, which is what I asked: the window is a lumping of the path-segment process and not of the state, the path-segment process of a Markov process is Markov, the two references are cited only for what they establish, and the refinement behaviour moves to the lemma with both branches explicit. Your added observation that under the first hypothesis the limit point has a property its approximants generically lack is a good one and is yours.

Findings 3, 4 and 5, the Wightman attribution, the Section 5 modest form, the Section 7 reconciliation with Section 3, and both positioning items are implemented; "finite resolution" is gone from Section 4 and "confirmed physics" from Section 7. Sara's two final points are implemented, and her formulation of the burden of proof is adopted with the two halves of the lemma as its two tests.

3. Lemma 1: what I verified, and where the statement falls short.

The proof is correct and I re-derived every step. The two hypotheses on δ are not decorative and the checking confirms it: $\delta \le 1/(2q^*)$ is exactly what gives $e^{-q^*\delta} \ge ½$, and $\delta \le (t-t_1)/2$ is exactly what places the lag $t-\delta-t_1$ inside the interval over which the minimum defining c is taken. The constant assembles as $2q^*\delta$ from the continuous part plus $4q^*\delta/c$ for each of the two conditional laws, giving $C = 2 + 8/c$. What supports the lemma is that re-derivation. I also ran stress tests, and I record them as such rather than as verification (four inequalities over 2,000 random generators with no violations, and a direct continuous-time simulation with Monte Carlo estimation for the rate; a test without violations does not prove a theorem).

The statement, however, does not cover the object it is about. It bounds the total variation "for every value y taken by $Y_\delta(t_1)$ with positive probability," and $Y_\delta(t_1)$ has a continuous component whenever a jump falls in the window, with probability of order $q^*\delta$, none of whose values has positive probability. So the statement controls the $|S|$ atomic rows and not the full conditional-kernel family. The asymmetry is visible in your own proof, which warns that sums over y′ (the intermediate time) are to be read as integrals where there is a continuous part, and does not make the same warning for the conditioning time t_1. The natural theorem is

ess sup over $y \sim L(Y_\delta(t_1))$ of $\|\Delta_\delta(\cdot|y)\|_{TV} \le C\, q^*\, \delta$,

with the proof rewritten in kernel integrals and regular conditional probabilities. The conclusion looks intact, and for a structural reason: your bound is $\max_x \Sigma_z |D_x(z)|$, uniform in x, so it survives any mixing over x and hence the disintegration. I checked numerically: conditioning on a band of the continuous part gives defects of the same order as the atomic rows and far below the bound. This is a formulation gap in the only new theorem of the paper, repairable in standard language without touching the argument (but it is not the theorem currently printed).

Two dependent claims need adjusting with it. First, the lemma proves $\le C\, q^*\delta$, which permits $o(\delta)$ and does not establish a linear leading term; the deposit asserts linear vanishing in three places, including the abstract. "Is $O(\delta)$", or "vanishes at least linearly", is what the proof delivers. You draw exactly this distinction, and correctly, in Theorem 5.1 of the convex-perturbation paper ("the rate is an upper rate… so no exact rate is asserted") so this is your own standard applied unevenly.
Second, non-lumpability implies failure of the Markov property but does not by itself imply a nonzero two-time CK defect, precisely because of the Feller example you cite. The hedge "generically positive" is right; the logical status should be stated, and persistence stated conditionally on the limiting lumped defect being nonzero.
My own counterexample makes the point inadvertently: I had to compute the defect separately, because the lumpability test does not deliver it.

4. Four further amendments.

Section 2 still presents the uniform interval average as the operational definition of the state; make it explicitly conditional on the observation model of equation (1). In Section 4, Λ names the physical validity scale in one passage and the sliding Wilsonian cutoff in another, with $\Lambda \to \infty$ for the continuum limit — two quantities under one symbol, in the section whose purpose is to separate scales; use two symbols, describe asymptotic freedom as one route to a UV completion rather than the exhaustive case, and do not identify ultraviolet divergence with finite physical validity. On the algebraic side, drop the clause discussed in item 1 and either state Wightman's theorem for bounded fields or add the domain hypotheses the unbounded case requires; I confirmed the bibliographic entry at source but not the hypothesis set you now print, and that check should be completed. Finally, keep the minimum-time language consistently semiclassical and modal: those arguments support an operational obstruction, not a universal theorem about every measurement protocol.

5. On the originality grade.

You are partly right and I am reformulating.

What I concede: a grade on two axes, one of which does not apply, invites a reader to average them, and the axis that applies here is the foundational one, where my grade was "moderate" and sat beside a priority finding that located no prior work assembling these three lines or proposing the criterion. The corrected report grades on that axis and states expressly that the paper is not offered as, and should not be weighed as, new mathematics or new physics. You say so yourself without reservation.

What I do not concede: the second axis was not gratuitous. The Institute's substantive standard for theoretical work is logical consistency with a testable hypothesis, so a foundational paper is weighed by whether its thesis does work, and doing work is measured in consequences. Your EPR comparison is admissible (EPR proved no theorem) but it is also the answer to your question. EPR's standing comes from what the argument licensed: a criterion that later became an experimentally decidable proposition distinguishing the hypothesis from its negation.

So, to answer you directly: yes, it is a result the thesis licenses and the presupposition forbids. Specifically, a consequence that follows from denying completeness and does *not* follow from a mere mis-specification of the state space. That is exactly what is missing. The one-sided test refutes the Markov property *on the observed state space*, as the paper says; every positive defect admits equally the explanation that the state space was badly chosen. The non-invasiveness criterion is imported and inherits its own assumption; it is a criterion for quantumness, not for incompleteness. What would raise the grade is a description-independent invariant, whose positive value cannot be absorbed by re-specifying the state space.

And here I must correct myself. I had written, in my own working record, that the later papers in the portfolio supply that invariant; the deficit with its intrinsic locus and β, the liftability profile and the memory rank. That is not established, and I have withdrawn it. Those quantities are invariants relative to a transition family and an observed space already fixed; they are not invariant under enlargement of the state space, change of observed variables, a different coarse-graining, hidden variables, or replacement of an effective description by an augmented one. My own report on the Koopman paper says as much (the deficit is relative to the chosen state description) and its Remark 2.8 records that unrestricted history augmentation always Markovizes.

The honest statement is therefore: the later work develops a much richer internal theory of the defect for a fixed observational description, and so strengthens the diagnostic this paper raises, but it does not remove the dependence on the choice of state space. Which means the gap is not a defect of this paper in particular. It is, on my present reading, the principal open problem of the portfolio as a whole, and identifying it may be worth more to you than the grade we have been arguing about. Paper 1's proper place is as the foundational premise of the series: moderate on its own, load-bearing in the set.

Subject to the amendments in items 3 and 4 (the first material, the rest focused or editorial) I regard the substantive questions of both reports as answered, withdrawn where appropriate, or correctly recorded as limitations and open problems.
Nothing outstanding requires new physics, and nothing disturbs the resolution of the two material findings. A corrected technical report will follow, carrying the five fixes in item 1.

Best,
Ezequiel